… To Operations Management in Manufacturing and Service Industries

Revision sheet

IN ONE SENTENCE

Operations management transforms **inputs** into **outputs**. You choose a **production method**, a **site** and a **capacity**, manage **purchasing, inventory and scheduling** (JIT, MRP, Gantt, PERT), adapt all this to **services**, ensure **quality** (TQM, SPC, ISO) and decide what to **outsource**.

TRANSFORMATION

Inputs
capital, HR, natural resources, information, materials, customers

→

Transformation
value added

→

Outputs
goods and services

THREE RESPONSIBILITIES

Production planning
how, where, which layout

Production control
daily monitoring, purchasing, inventory

Quality control
meeting specifications

PRODUCTION METHODS

Method	Key	Example
Make-to-order	to customer specs, low volume	Print shop, Burger King
Make-to-stock	high volume, low cost, inventory	Cars, McDonald's
Mass customization	volume + customization	Nike By You, Levi's, M&M's

SITE AND CAPACITY

- Site: near suppliers/customers, skilled workers, living conditions, low costs, business climate.
- PowerSki (San Clemente): suppliers, engineers, living conditions.
- **Capacity**: maximum output under normal conditions. Too low → lost sales; too high → waste.

MATERIALS MANAGEMENT

Purchasing
≈ 50% of costs; e-purchasing, EDI

JIT
delivery just as materials enter production

MRP
software calculates quantities and order dates

MRP II
MRP coordinated with other functions

MPS
what, how many, when; routing

GANTT AND PERT

Gantt
status of a project

PERT
activities, times, sequence; critical path

Critical path
the longest sequence

Hiking bear
10 + 30 + 15 + 5 + 5 = 65 min

LAYOUTS AND TECHNOLOGY (NOT FOR THE MCQ)

- Process, product (assembly line), cellular, fixed-position.
- CAD, CAM, CIM, FMS; automation, robots, 3D printing.

SERVICES

Weight

≈ 80% of U.S. jobs; 13% in goods

Differences

intangibility, customization, customer contact

Burger King

make-to-order (30 s) vs McDonald's make-to-stock (5 min)

BK site

traffic count: "through the light and to the right"

Capacity

services can't be stored; you schedule workers

QUALITY

Quality (ASQ)

characteristics that satisfy stated or implied needs

TQM

customers, employees (quality circles), continuous improvement

SPC

samples and control chart (Raisin Bran)

Benchmarking

L.L. Bean, 3M, Motorola, Mary Kay

ISO

9000 quality, 14000 environment; voluntary

OUTSOURCING

- Having **outside vendors** make products.
- PowerSki: assembly at Experience Powersports (Moses Lake), components to its specifications.
- Reasons: expertise, labor costs, focus on the core business.

TRAPS

- BK = **make-to-order**; McDonald's = **make-to-stock**.
- MRP = materials; MRP II = the whole company.
- Shortening an activity **off** the critical path does not reduce total time.
- ISO focuses on **how work is done**, not on the product.